

Digital Electronics

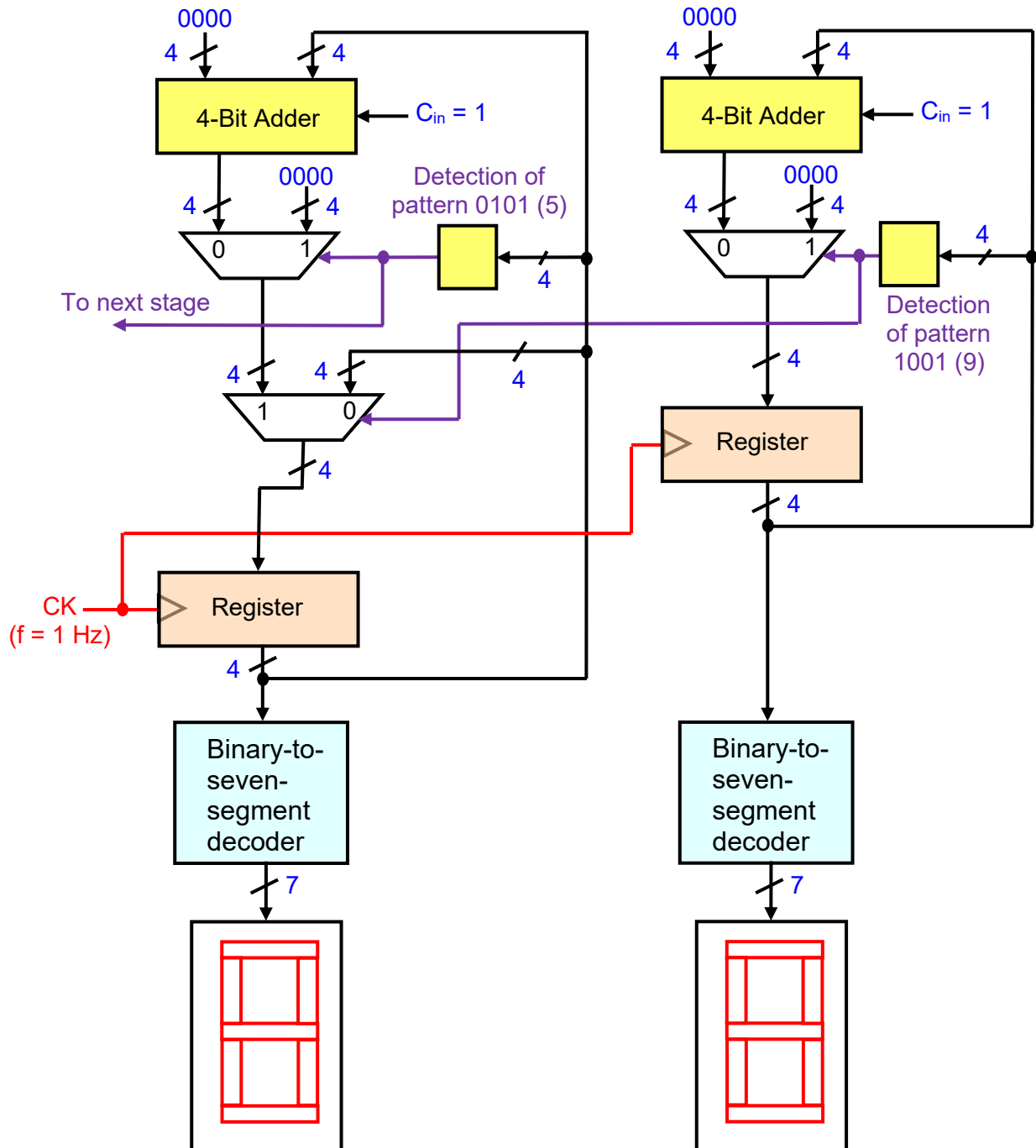
Circuit Example – A Stopwatch

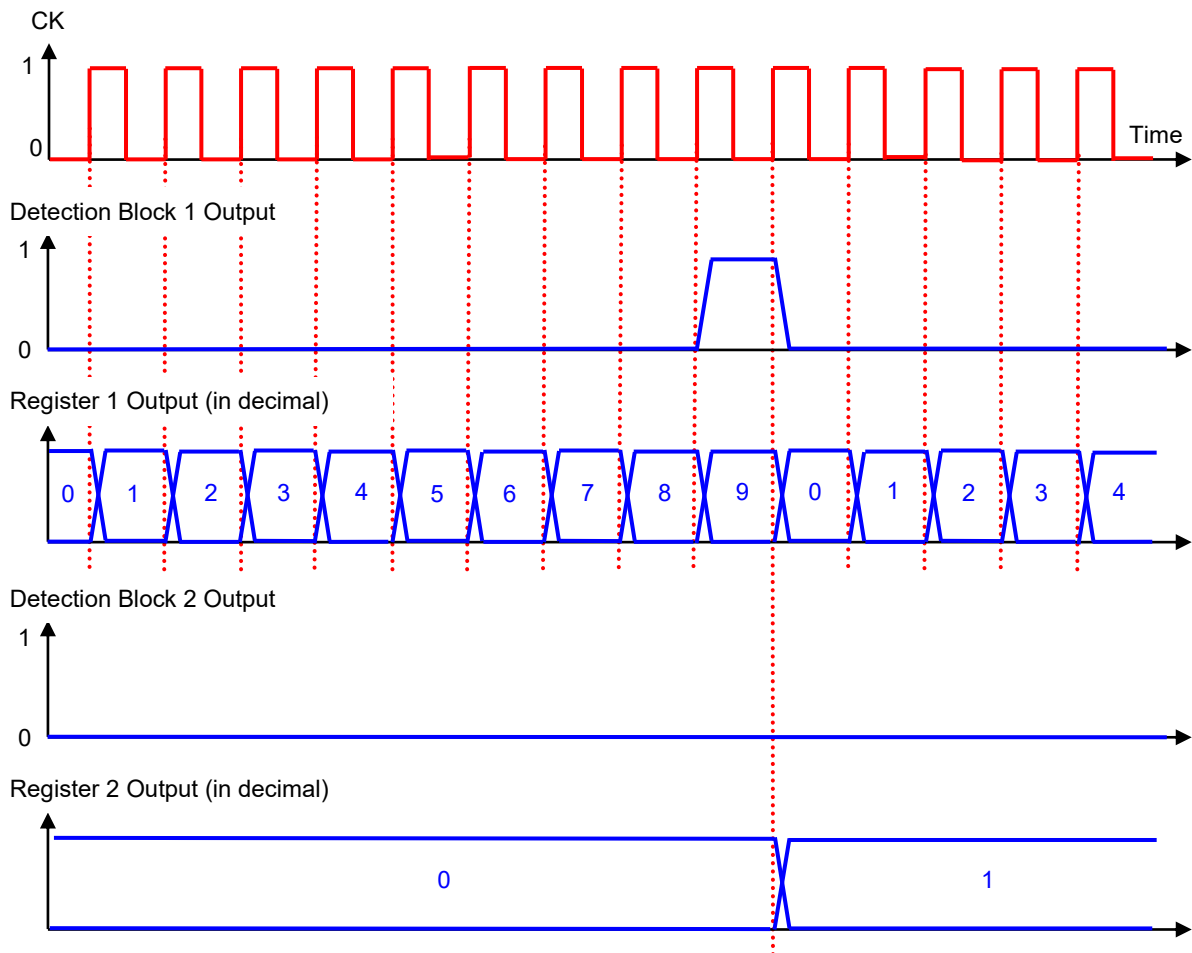
S Le Goff

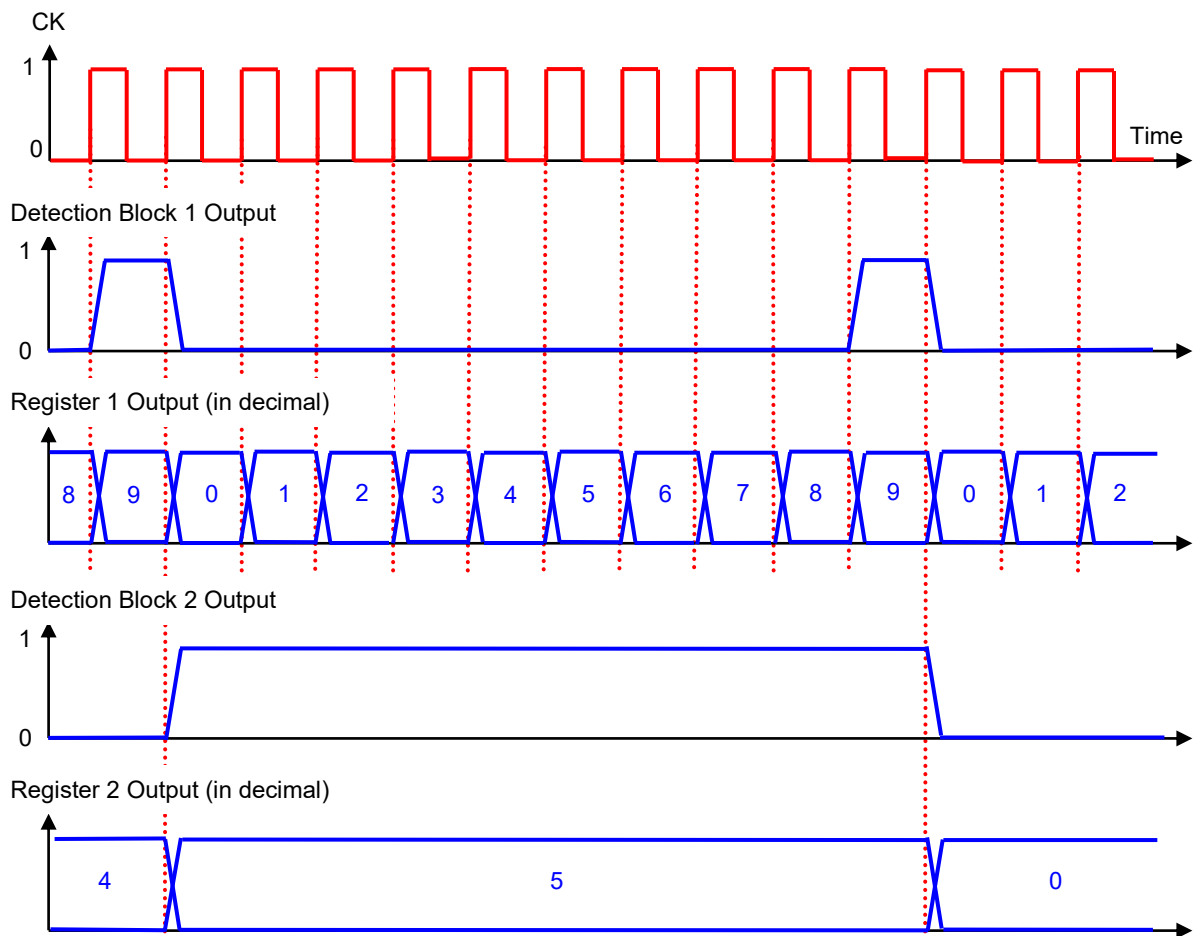
June 2026

3. Generic stopwatch

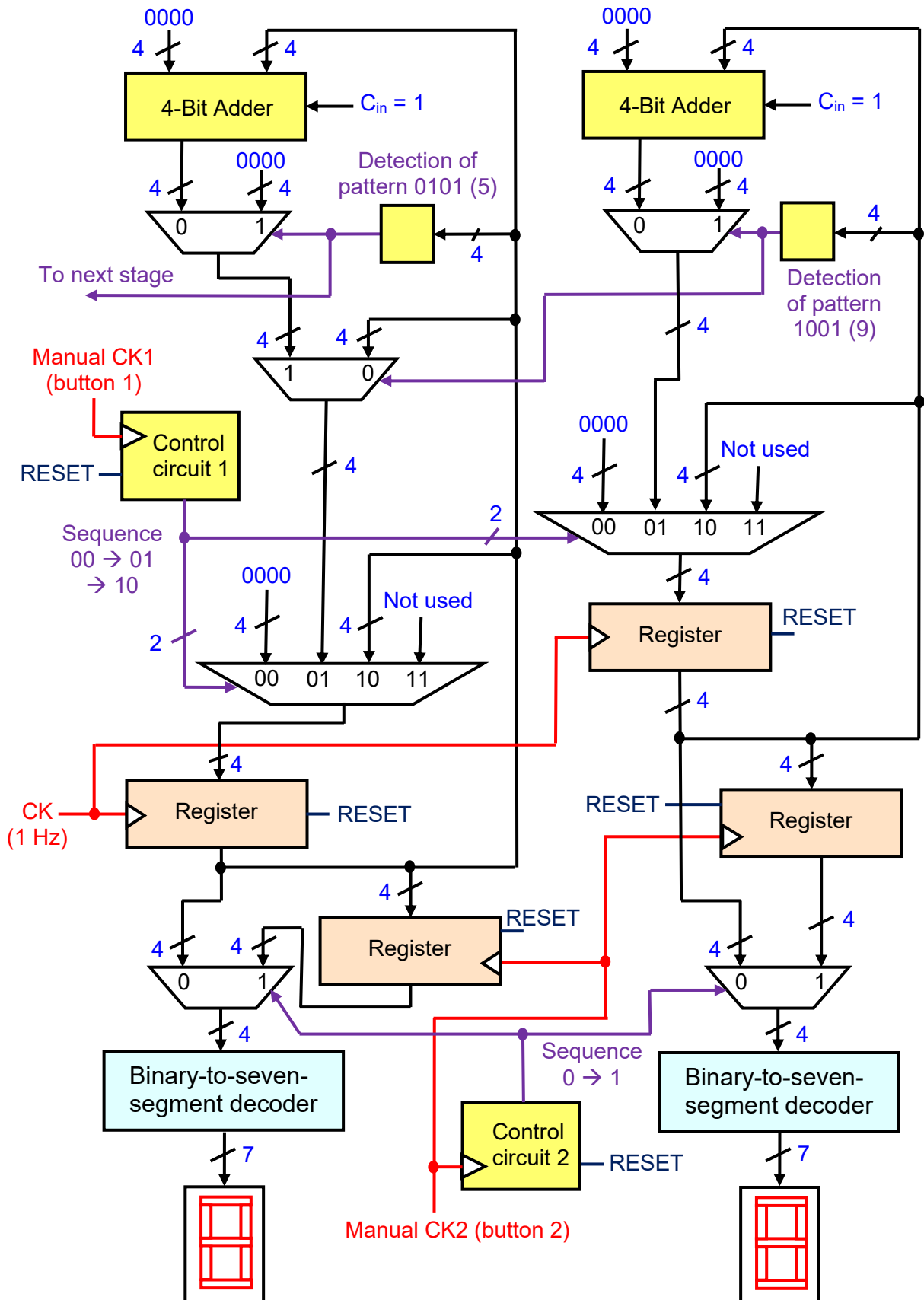
Design of a stopwatch circuit that runs from 0 to 59 seconds.







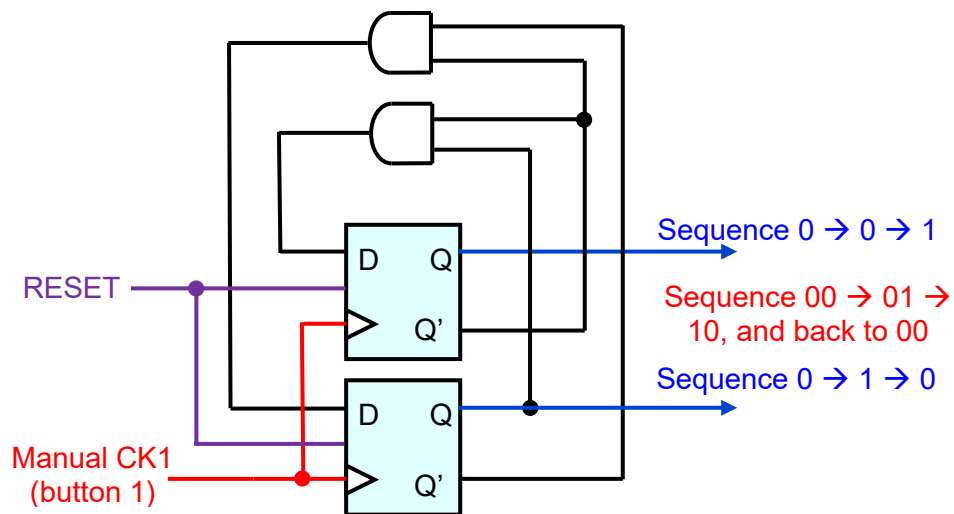
4. Improved stopwatch - Practical stopwatch circuit that can run from 0 to 59 seconds.



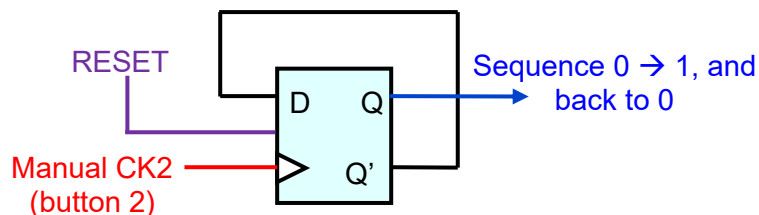
Press button 1 to start the counting, press it once again to stop the counting, and press it a third time to reset the stopwatch.

Press button 2 to freeze the 7-segment displays without interrupting the counting, and press it once again to unfreeze the displays.

Control circuit 1:



Control circuit 2:



As soon as the power supply is switched on, the RESET signal must be active, e.g. equal to one, in order to ensure that the content of all registers is initially equal to zero. Then, this signal must quickly fall to zero in order to become inactive as long as the power supply remains on.

The RESET signal is thus a short-duration pulse that can be generated using a simple circuit such as the one depicted below.

